Mehmet Aygun

https://mehmetaygun.github.io/

Email : mayguntr@gmail.com
Mobile : +44-739-845-8684
Github : github.com/MehmetAygun

Education

- **The University of Edinburgh (UoE)** Edinburgh, UK
  Doctor of Philosophy in Computer Science *Sep. 2021 – Current*
- **Technical University of Munich (TUM)** Munich, Germany
  Master of Science in Computer Science *Oct. 2018 – Dec. 2020*
- **Istanbul Technical University (ITU)** Istanbul, Turkey
  Bachelor of Science in Computer Science *Sep. 2013 – June. 2017*

Research Experience

- **Institute for Adaptive and Neural Computation at UoE** Edinburgh, UK
  Postgraduate Research Student *Sep 2021 - Current*
    - **3D Learning**: Working on shape, semantics and learning triangle. Advisors: Dr. Oisin Mac Aodha
- **TUM Computer Vision Lab** Munich, DE
  Research Student *Oct 2018 - Jun 2021*
    - **Holistic Scene Analysis**: Worked on semantic/instance segmentation and multi object tracking with Lidar data. Advisors: Prof. Laura Leal-Taixe, Dr. Aljosa Osep.
    - **3D Shape Analysis**: Worked on shape correspondence problem with self-supervised deep learning methods using functional maps and heat kernels. Advisors: Zorah Lahner, Prof. Daniel Cremers.
- **ITU Computer Vision Lab** Istanbul, Turkey
  Research Student *Dec 2013 - June 2017*
    - **Medical Image Segmentation**: Worked on multi-modal 3D Brain Tumor segmentation using convolutional neural networks under supervision of Prof. Gozde Unal.
    - **Transfer Learning**: Worked on transfer learning for convolutional neural networks. Collaborated with Dr. Yusuf Aytar from MIT CSAIL and proposed a statistical regularization method for transferring knowledge between different CNN architectures.

Publications

- **Conference Papers**:
    - **M. Aygun**, A. Osep, M.Weber, M.Maximov, C. Stachniss, J. Behley, L. Lael-Taixe, ”4D Panoptic Lidar Segmentation”, In IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021
    - **M. Aygun**, Z. Lahner, D. Cremers, ”Unsupervised Dense Shape Correspondence using Heat Kernels”, In International Conference on 3D Vision (3DV), 2020
- **Workshop Papers & Preprints**:
    - **M. Aygun**, Y. Aytar, H. K. Ekenel, ”Exploiting Convolution Filter Patterns for Transfer Learning”, ICCV TASK-CV workshop, 2017
    - R. C. Malli*, **M. Aygun***, H. K. Ekenel, ”Apparent Age Estimation Using Ensemble of Deep Learning Models”, CVPRW, June 2016 (*:equal contribution.)
    - **M. Aygun**, Y. Sahin, G. Unal, ”Multi modal Convolutional Neural Networks for Brain Tumor Segmentation, arXiv preprint arXiv:1809.06191 (2018)

Honors & Awards

- **ICCV 17 Workshop, Honourable Mention Award**: Received Honourable Mention Award at ICCV 17 TASK-CV workshop for our paper 'Exploiting Convolution Filter Patterns for Transfer Learning'.
- **DAAD scholarship**: 2 year master scholarship for studying in Germany.
- **Bachelor Thesis Award**: Best Bachelor Thesis Award from Computer Engineering Department for my thesis titled ”Multi Modal 3D Convolutional Neural Networks for Brain Tumor Segmentation”
- **Undergraduate Research Scholarship** : Research scholarship from science and technology research council of Turkey during my last two year of bachelor education.

Professional Experience

- **Coriolis Pharma** Munich, Germany
  Computer Vision Engineer *Sep 2019 - March 2020*
  - **Biological image analysis**: Developed methods for particle analysis using deep learning based classification, instance segmentation and unsupervised learning methods. Used frameworks: Tensorflow, Keras, OpenCV.
- **Motius** Munich, Germany
  Machine Learning Engineer *Jan 2019 - April 2019*
  - **Handwritten Text Recognition**: Developed handwritten text recognition system using deep learning (CNN+LSTM+CTC). Reduced word error rate from 18% to 12%. Used frameworks: Tensorflow, OpenCV.
- **Bunsar** Istanbul, Turkey
  Computer Vision Engineer *Feb 2017 - Sep 2018*
  - **Visual Search Engine from Images and Videos**: Research and development activities for image retrieval system using deep learning methods. Topic studied: Object Detection, Image Segmentation, Image Classification, Large Scale Image Retrieval Used frameworks: Caffe, OpenCV, Sklearn

Skills

- **Programming**: Python, C/C++, Matlab
- **Frameworks and Tools**: Pytorch, Tensorflow, Caffe, OpenCV, Sklearn, Pandas, SQL, Numpy, Vim, Git, Linux.

Related Coursework

- **TUM**: Natural Language Processing, Deep Learning, Variational Methods, Multiple View Geometry, Tracking and Detection
- **ITU**: Computer Vision, Learning from Data, Probability, Linear Algebra, Calculus I & II

Other

- **Reviewer**: Reviewed 8 papers for MICCAI Deep Learning Workshop, 2017-2018
- **Volunteer**: Volunteering work for high school student at code.org

References

- **Prof. Laura Leal-Taixe**: leal.taixe@tum.de
- **Prof. Gozde Unal**: gozde.unal@itu.edu.tr
- **Prof. Hazim Kemal Ekenel**: ekenel@itu.edu.tr